

Step 1:

Field	Value
Step	1
Module	RECON
Hypothesis	Identify running services and application stack on port 8081
Action	nmap -sV -sC -p 8081 100.113.75.28
Outcome	Success
Pivot?	No
Notes	Apache 2.4.65 (Debian), PHP app (DVWA). robots.txt exposes /admin/, /backup/, /debug/

Step 2:

Field	Value
Step	2
Module	ENUMERATION
Hypothesis	robots.txt paths may expose hidden content
Action	curl robots.txt and request /admin/, /backup/, /debug/
Outcome	Fail
Pivot?	Yes
Notes	All returned 404. Paths likely removed or non-existent

Step 3:

Field	Value
Step	3
Module	ENUMERATION
Hypothesis	Directory brute-force may reveal hidden files or endpoints
Action	gobuster dir -u http://100.113.75.28:8081/ -w common.txt -x php,txt,html
Outcome	Success
Pivot?	Yes
Notes	flag.txt discovered at web root (HTTP 200)

Step 4:

Field	Value
Step	4
Module	FILE_DISCLOSURE
Hypothesis	flag.txt is directly accessible and contains the flag
Action	curl http://100.113.75.28:8081/flag.txt
Outcome	Success
Pivot?	No
Notes	FLAG{INITIAL_FOOTHOLD_ACHIEVED} retrieved via direct HTTP GET

Metrics:

Total steps: 4

Total pivots: 3

Dead-end modules: 1

Exploit chain depth: 2

Branching factor: total pivots/ total steps = $3 / 4 = \mathbf{0.75}$